

Primary Key vs UNIQUE + NOT NULL

A simple Oracle explanation with examples

Main idea

A PRIMARY KEY is the official identifier of each row. UNIQUE + NOT NULL also prevents duplicates and nulls, but it is usually used for other important columns, called alternate keys.

PRIMARY KEY = UNIQUE + NOT NULL + Main row identity

Simple comparison

Feature	PRIMARY KEY	UNIQUE + NOT NULL
Purpose	Main identifier for every row	Prevents duplicate values and nulls
How many per table?	Only one PRIMARY KEY constraint	Many UNIQUE + NOT NULL constraints are allowed
Null values	Never allowed	Not allowed because of NOT NULL
Duplicate values	Never allowed	Never allowed because of UNIQUE
Typical columns	employee_id, order_id, student_id	email, username, passport_number
Meaning	This is the row identity	This is another value that must stay unique

Example 1: Use both in the same table

This is the most common design. The ID is the primary key, while the email is also required to be unique.

```
CREATE TABLE employees (
  employee_id NUMBER
    CONSTRAINT employees_pk PRIMARY KEY,

  employee_name VARCHAR2(100) NOT NULL,

  email VARCHAR2(150)
    CONSTRAINT employees_email_nn NOT NULL
    CONSTRAINT employees_email_uk UNIQUE
);
```

- employee_id identifies the employee row.
- email cannot be NULL.
- Two employees cannot have the same email.
- The table still has only one primary key: employee_id.

Testing the constraints

```
-- Accepted
INSERT INTO employees VALUES (1, 'Ali', 'ali@example.com');
INSERT INTO employees VALUES (2, 'Mona', 'mona@example.com');

-- Rejected: duplicate PRIMARY KEY value
INSERT INTO employees VALUES (1, 'Omar', 'omar@example.com');

-- Rejected: duplicate UNIQUE email
INSERT INTO employees VALUES (3, 'Sara', 'ali@example.com');

-- Rejected: email is NOT NULL
INSERT INTO employees VALUES (4, 'Nour', NULL);
```

Result

Both constraints reject duplicates and nulls. The difference is their role: employee_id is the official row identity; email is an additional unique business value.

Example 2: UNIQUE without NOT NULL

This is different. In Oracle, a UNIQUE column can contain multiple NULL values unless you also add NOT NULL.

```
CREATE TABLE users_demo (  
    user_id NUMBER PRIMARY KEY,  
    nickname VARCHAR2(50) UNIQUE  
);  
  
-- Both rows are accepted because nickname is NULL  
INSERT INTO users_demo VALUES (1, NULL);  
INSERT INTO users_demo VALUES (2, NULL);
```

Important

UNIQUE alone does not mean that a value is required. Add NOT NULL when the column must always contain a value.

Can UNIQUE + NOT NULL replace a primary key?

For a single column, the data rules are almost the same: **no nulls and no duplicates**.

- However, Oracle metadata still knows whether a constraint is PRIMARY KEY or UNIQUE.
- A table may have only one primary key, but it may have several unique constraints.
- Developers and database tools expect the primary key to represent the main row identity.
- Foreign keys can reference a primary key or an appropriate unique key, but the primary key is normally the default parent identifier.

Which one should you use?

Use PRIMARY KEY when...	Use UNIQUE + NOT NULL when...
- The column is the main ID of the row. - Other tables will normally refer to this row by that value. - Example: employee_id or order_id.	- The column is important and must be unique, but it is not the main ID. - The table already has a primary key. - Example: email or username.

Easy rule to remember

Every table should normally have one PRIMARY KEY. Add UNIQUE + NOT NULL to other columns that must also be present and never repeated.

Final summary

- PRIMARY KEY: the main identity of the row.
- UNIQUE: no duplicate non-null values.
- NOT NULL: a value is required.
- UNIQUE + NOT NULL: no duplicates and no missing values, but it is still not automatically the table primary key.

Best common design: ID as PRIMARY KEY, email as UNIQUE + NOT NULL.